Magnetic Circuits and Electrical Analogies

1 Magnetic Circuits and Their Electrical Analogies

Placing a ferromagnetic material in the path of magnetic flux lines increases flux density as it offers a low-reluctance path.

Permeability and Magnetic Circuits

- **High Permeability** (μ): Corresponds to Low Reluctance
- A **magnetic circuit** guides magnetic fields along paths with minimal reluctance.

$$\text{MMF} \equiv \text{EMF (analogous source)} \Rightarrow F = \Phi\mathcal{S}$$

$$\text{Reluctance } (\mathcal{S}) = \frac{l}{\mu A} \quad \text{where: } \begin{cases} l = \text{length of magnetic path} \\ A = \text{cross-sectional area} \\ \mu = \text{permeability} \end{cases}$$

$$F = NI \quad \text{(Number of turns } N \text{ times current } I\text{)}$$

Magnetic Quantities and Analogies

- Φ (phi): Magnetic Flux [Weber, Wb]
- B: Flux Density $= \frac{\Phi}{A}$ [Tesla]
- H: Magnetic Field Intensity $= \frac{F}{l}$ or $B = \mu H$

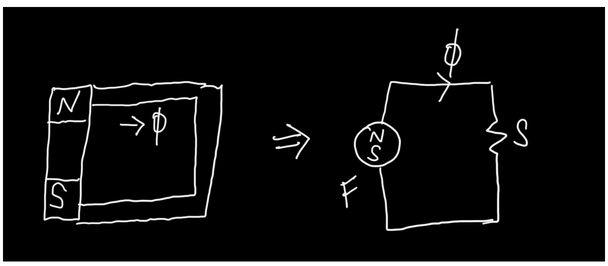

Figure 1:

Magnetic vs. Electrical Analogy

Magnetic	Electrical
Flux (Φ) - Wb	Current (I) - A
MMF (F) - Ampere-Turns	EMF/Voltage (V) - Volts
Reluctance ($\mathcal{S}$)	Resistance (R)
Flux Density (B) - Tesla	Current Density

Key Equations

$$F = NI$$
$$H = \frac{F}{l}$$
$$F = \Phi\mathcal{S}$$
$$\mathcal{S} = \frac{l}{\mu A}$$
$$B = \mu_0\mu_r H$$

Solved Example 1

Given:

$$A = 10\,\text{cm}^2 = 10 \times 10^{-4}\,\text{m}^2$$
$$N = 200$$
$$i = 3\,\text{A}$$
$$\mu_r = 1200$$
$$l_{\text{core}} = 25\,\text{cm} = 0.25\,\text{m}$$
$$l_{\text{air}} = 1\,\text{mm} = 10^{-3}\,\text{m}$$
$$\mu_0 = 4\pi \times 10^{-7}\,\text{H/m}$$

Objective: Find B
Theory:

$$F = \Phi\mathcal{R} = F_{\text{core}} + F_{\text{air}} = B\left(\frac{l_{\text{core}}}{\mu_0\mu_r} + \frac{l_{\text{air}}}{\mu_0}\right)$$

$$\text{MMF} = Ni = 200 \times 3 = 600$$
$$600 = B\left(\frac{0.25}{1200\mu_0} + \frac{10^{-3}}{\mu_0}\right)$$
$$600 = B\left(\frac{0.25 + 1200 \times 10^{-3}}{1200\mu_0}\right)$$
$$600 = B\left(\frac{1.45}{\mu_0}\right) \Rightarrow B = \frac{600\mu_0}{1.45}$$
$$B \approx 0.58\,\text{T}$$

Solved Example 2

Given:

$$
\begin{aligned}
N &= 500 \\
\text{Average Diameter} &= 30\,\text{cm} \Rightarrow l = \pi D = \pi \times 0.3\,\text{m} \\
\Phi &= 4\,\text{mWb} = 4 \times 10^{-3}\,\text{Wb} \\
\mu_r &= 800 \\
\mu_0 &= 4\pi \times 10^{-7}
\end{aligned}
$$

Objective: Find current i

$$Ni = \frac{\Phi l}{\mu_0 \mu_r A}$$

Where:

$$
\begin{aligned}
A &= 1\,\text{cm}^2 = 1 \times 10^{-4}\,\text{m}^2 \\
l &= \pi \times 0.3
\end{aligned}
$$

$$
\begin{aligned}
i &= \frac{\Phi \cdot l}{\mu_0 \mu_r N A} = \frac{4 \times 10^{-3} \cdot \pi \cdot 0.3}{4\pi \times 10^{-7} \cdot 800 \cdot 500 \cdot 10^{-4}} \\
i &\approx 1.1\,\text{A}
\end{aligned}
$$

Solved Example 3

Given:

$$
\begin{aligned}
N &= 500 \\
B &= 0.8\,\text{T} \\
d &= 30\,\text{cm} \Rightarrow r = 15\,\text{cm} = 0.15\,\text{m} \\
\mu_r &= 800
\end{aligned}
$$

Objective: Find i

$$R = \frac{l}{\mu A} = \frac{2\pi r}{\mu_0 \mu_r A}$$

Area:

$$A = 6.25 \times 10^{-4}\,\text{m}^2, \quad \Phi = B \cdot A = 0.8 \cdot 6.25 \times 10^{-4} = 5 \times 10^{-4}\,\text{Wb}$$

$$Ni = \Phi \cdot \mathcal{R} = \frac{5 \times 10^{-4} \cdot 2\pi \cdot 0.15}{\mu_0 \mu_r A} \Rightarrow i \approx 0.23\,\text{A}$$

Solved Example 4

Magnetic Circuit with three branches.
Given:

$$N = 300, \quad \Phi = 0.36\,\text{Wb (at center)}, \quad R_1 = \frac{(6+6+6)\times 10^{-2}}{\mu A}, \quad R_2 = \frac{6\times 10^{-2}}{\mu A}, \quad R_3 = \frac{18\times 10^{-2}}{\mu A}$$

$$\Phi_3 = \frac{6\times 10^{-3}}{18\times 10^{-3}}\cdot \Phi = 0.12, \quad \Phi_1 = 0.48$$

$$Ni = \frac{0.48\cdot 0.18 + 0.18\cdot 0.06}{\mu A} \Rightarrow i = \frac{F}{N} = \frac{\text{Total MMF}}{300} \approx 0.16\,\text{A}$$

2 Basic Concepts of Electromagnetic Induction

- Moving charges cause magnetic flux (ϕ)
- By Faraday's Law: Moving flux induces voltage
- If v is DC $\Rightarrow$ i is DC $\Rightarrow$ constant ϕ

$$\Rightarrow \text{No emf induced}$$

- If v is AC $\Rightarrow$ i is AC $\Rightarrow$ time-varying ϕ $\Rightarrow$ induced emf:

$$v = N\frac{d\phi}{dt}$$

- Magnetic flux from current:

$$\phi = \frac{Ni}{R}$$

- Substituting in Faraday's Law:

$$v = N\frac{d\phi}{dt} = \frac{N^2}{R}\frac{di}{dt}$$

3 Inductance and Magnetic Analogy

- Inductance is defined as:

$$L = N\frac{d\phi}{di}$$

- From $\phi = \frac{Ni}{R}$:

$$L = \frac{N^2}{R}$$

- This shows magnetic reluctance R is analogous to resistance in electrical circuits.

4 Transformers and Mutual Inductance

4.1 Transformer Principle

- Voltage induced in secondary:
$$v_2 = N_2 \frac{d\phi}{dt}$$
- Mutual inductance is the coupling factor:
$$M = k\sqrt{L_1 L_2}$$
- Turns ratio governs voltage transformation:
$$\frac{v_2}{v_1} = \frac{N_2}{N_1}$$
- Transformers reject DC since a constant ϕ does not induce emf
- v_1 is known as primary voltage and v_2 is known as secondary voltage.
- If $\frac{N_2}{N_1} > 1$, it is called a step-up transformer, and a step-down transformer if $\frac{N_2}{N_1} < 1$.

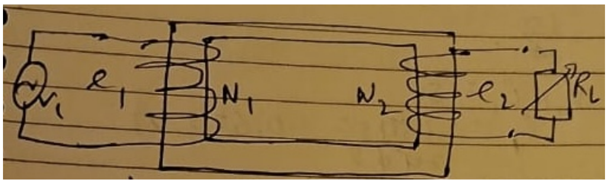

Figure 2:

4.2 Power and Current Relations in Ideal Transformers

$$v_1 i_1 = v_2 i_2 \Rightarrow i_1 = \frac{v_2 i_2}{v_1}$$

5 Worked Example 1: Charger Load on Step-down Transformer

- Given:
$$v_1 = 240\text{V (rms)}, \quad N_1 = 1000, \quad N_2 = 100$$
- Step-down transformer, turns ratio $\frac{N_2}{N_1} = \frac{1}{10}$
- Load: $P = 20\text{W}$ charger

Find: Secondary voltage and current.
Using the relation

$$\frac{v_2}{v_1} = \frac{N_2}{N_1}$$

$$v_2 = 24\text{V}, \quad i_2 = \frac{20}{24} = 0.83\text{A}$$

Primary current: Since we consider the transformer as ideal, there will be no power loss.

$$i_1 = \frac{v_2 i_2}{v_1} = \frac{24 \times 0.83}{240} = 0.083\text{A}$$

6 Worked Example 2: Transformation Ratio and Input Current

Given: A 240V transformer supplies power to a 12V, 150W lamp.

- Transformation ratio:
$$\frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{12}{240} = 0.05$$
- Output current:
$$i_2 = \frac{150}{12} = 12.5\text{A}$$
- Input current:
$$i_1 = \frac{150}{240} = 0.625\text{A}$$

7 Worked Example 3: 5kVA Transformer Load Analysis

Given:

- Transformer rating = 5kVA
- $\frac{N_2}{N_1} = 0.1$
- $V_1 = 2500\text{V}$ supply

(a) Find secondary (load) current

$$P = 5000\text{W}, \quad V_2 = 250\text{V} \Rightarrow i_2 = \frac{5000}{250} = 20\text{A}$$

(b) Minimum Load Resistance

$$P_L = \frac{V^2}{R_L} \Rightarrow R_L = \frac{V^2}{P_L} = \frac{250^2}{5000} = 12.5\ \Omega$$

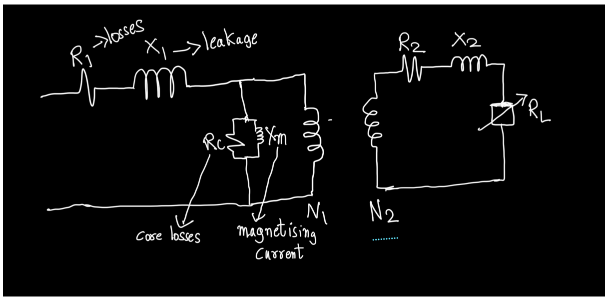

Figure 3:

Transformer Equivalent Circuit

- Includes:
  - Winding resistances R_1, R_2
  - Magnetizing reactance X_m
  - Core losses
  - Load R_L
- Core losses include Hysteresis losses and Eddy Current losses.

Contribution Statement

Saket Sharma: Editing and Compilation of notes
Vipulav Batra: taken notes for 3rd lecture
Varanasi Sumant: taken notes for 1, 2
K S Kevin: diagrams and examples